INT396 - Unsupervised Learning | Section K3P24MK | CA-1 Groups and Projects

Find your group by your name or registration number. The objectives are what your project will be assessed on; the advanced extension is optional. Use the official source to read the original problem statement (for Smart India Hackathon problems, search the PS ID on the linked page).

Group	Students	Project Title	Problem to Solve	Who Benefits	Objectives (what you will be assessed on)	Suggested Techniques	Syllabus Topics Used	Data to Use	Optional Advanced Extension	Inspired By (official source)
G1	Mohammad Raza (12400651)	**Label-Free Threat Detection in One-Way Network Traffic**	Critical infrastructure is monitored through one-way taps where operators see everything crossing the link but have no labelled catalogue of what an attack looks like.	Critical infrastructure operators, security operations centres	1. Build a detector that models normal one-way traffic and surfaces departures without any attack labels. 2. Produce a label-free detector evaluated on attack classes held out entirely from training. 3. Report detection rate on unseen attack classes, false positives per hour, PR-AUC. Baseline: supervised classifier on known attacks only, mandatory. Deliver SOC dashboard with traffic timeline and anomaly bands.	Unsupervised anomaly detection on traffic features; per-link baselines	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	CICIDS2017, UNSW-NB15, NSL-KDD (public) (Find and cite suitable public data yourselves.)	Cluster the flagged traffic to separate distinct attack families.	SMART INDIA HACKATHON 2026 PS ID: SIH26145 AI-Based Detection of Cyber Threats in Unidirectional IP Traffic Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G2	Rini Verma (12409359)	**Log Anomaly Discovery Across Heterogeneous Enterprise Sources**	Enterprises generate logs from dozens of incompatible sources, and failures announce themselves in patterns nobody reads until an outage.	Enterprise IT operations, incident response teams	1. Build a detector that normalises heterogeneous logs into a common representation and flags abnormal behaviour without labels. 2. Produce a log anomaly detector reporting warning time before each recorded incident. 3. Report lead time before incident, false alarms per thousand lines, precision at top-K. Baseline: keyword error-rate threshold, mandatory. Deliver AIOps dashboard with ranked anomalous windows.	Log parsing and templating; unsupervised anomaly scoring on event sequences	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit VI association and sequence mining	HDFS, BGL and Thunderbird log datasets (public) (Find and cite suitable public data yourselves.)	Add per-source baselines so a noisy service does not dominate the alerts.	SMART INDIA HACKATHON 2026 PS ID: SIH26156 Universal Log Pre-processing Framework Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G3	Palakpreet Kaur (12408423)	**Alert Triage by Unsupervised Scoring in a Security Operations Centre**	Analysts receive far more alerts than they can read, so genuine incidents wait behind noise and alert fatigue sets in.	SOC analysts, supervisory security managers	1. Build a triage system that scores alerts by how unusual they are and reorders the analyst queue. 2. Produce an alert ranking with precision at top-K against recorded incidents and the review effort saved. 3. Report precision at top-K, incident recall in the reviewed fraction, alerts avoided. Baseline: chronological queue, mandatory. Deliver SOC triage dashboard with ranked alert queue.	Unsupervised anomaly scoring; ranking for review; workload measurement	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	CICIDS alert logs and public SIEM datasets (Find and cite suitable public data yourselves.)	Report how many true incidents fall outside the reviewed fraction.	SMART INDIA HACKATHON 2026 PS ID: SIH26157 Supervisory Analytics Tool for SOC Assessment (SAT-SA) Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G4	Snawarpreet Singh (12417951) Naina Vaid (12415429)	**Unusual Pattern Discovery in Bitcoin Transaction Traffic**	Cryptocurrency transaction volume is far beyond manual review and illicit activity carries no label until an investigation is already underway.	Financial intelligence units, blockchain analytics teams	1. Build a detector that models normal transaction behaviour and surfaces statistically unusual flows for investigator review. 2. Produce a ranked anomaly list with precision on a manually verified sample. 3. Report precision at top-K on verified sample, flag rate, PR-AUC where labels exist. Baseline: transaction-amount threshold, mandatory. Deliver Investigator dashboard with ranked flows.	Unsupervised anomaly detection on transaction features; graph-derived features	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Elliptic Bitcoin dataset (public) (Find and cite suitable public data yourselves.)	Add wallet-level aggregation so patterns spanning many transactions are visible.	SMART INDIA HACKATHON 2026 PS ID: SIH26146 AI-Powered Monitoring & Analysis of Bitcoin Transaction Traffic Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G5	Samrath Thukral (12419409) S Selva Ganesh (12409492)	**Fraud-Linked Wallet Discovery from Victim Reports**	Victims report suspect wallet addresses, but the layering wallets between the victim and the exchange are never enumerated.	Cybercrime investigation units, victims of financial fraud	1. Build a system that scores wallets by how far their behaviour departs from ordinary use and surfaces likely layering chains. 2. Produce a wallet anomaly ranking with tracing chains, validated on reported cases. 3. Report precision at top-K, chain recall on reported cases, false positives per thousand wallets. Baseline: direct-recipient-only tracing, mandatory. Deliver Investigation dashboard with wallet chains.	Unsupervised anomaly scoring on wallet behaviour; sequence tracing	Unit V anomaly detection; Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering	Elliptic Bitcoin dataset and public blockchain extracts (Find and cite suitable public data yourselves.)	Report how many hops the tracing survives before the signal is lost.	SMART INDIA HACKATHON 2026 PS ID: SIH26183 Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G6	Bhanu Sesha Sai Chalamalasetti (12400638)	**Email Threat Discovery Without Labelled Phishing Examples**	Email remains the most exploited attack channel, and each new campaign is unlike anything already in the block list.	Enterprise and government email administrators, end users	1. Build a detector that models normal mail for each organisation and surfaces messages that depart from it. 2. Produce a label-free detector evaluated against held-out phishing labels with a stated false-positive rate. 3. Report detection rate on held-out phishing, false positives per thousand legitimate mails, PR-AUC. Baseline: keyword and blocklist rules, mandatory. Deliver Mail security dashboard with flagged message queue.	Unsupervised anomaly detection on header and content features; per-domain baselines	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Enron-Spam, SpamAssassin and PhishTank corpora (public) (Find and cite suitable public data yourselves.)	Cluster flagged mail to separate campaigns from one-off attempts.	SMART INDIA HACKATHON 2026 PS ID: SIH26106 AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform Organisation: All India Council for Technical Education (AICTE) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G7	Jeetaksh Luhaniwal (12411743)	**Sensor Fault Discovery in Automatic Weather Station Networks**	Weather stations feed forecasting and disaster response continuously, and a silently faulty sensor corrupts every downstream decision.	Meteorological departments, disaster management authorities	1. Build a detector that separates a genuine environmental extreme from a failing instrument, without labelled failures. 2. Produce a detector reporting accuracy separately for sensor faults and for genuine extremes. 3. Report fault detection rate, extreme-event false-flag rate, precision at top-K. Baseline: fixed range checks, mandatory. Deliver Network health dashboard with per-station status.	Unsupervised anomaly detection on multivariate sensor streams; cross-station agreement	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Intel Berkeley Lab sensor data; CPCB and OpenAQ station series (public) (Find and cite suitable public data yourselves.)	Use neighbouring stations to arbitrate ambiguous cases.	SMART INDIA HACKATHON 2026 PS ID: SIH26073 AI/ML-Based Intelligent Anomaly Detection for Automatic Weather Stations (AWS) Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G8	Prakhar Mohan (12415288) Vivek Poonia (12416020)	**Site-Specific Baselines for a Resilient Environmental Monitoring Network**	Monitoring sites have genuinely different normal conditions, so one global threshold buries the network in alarms from naturally active sites.	Environmental agencies, district disaster cells	1. Build a detector that learns each site own normal and scores departures locally rather than against a global rule. 2. Produce a local-baseline detector with alarm burden compared against global thresholding at matched detection. 3. Report alarm count at matched detection rate, precision at top-K, per-site false alarms. Baseline: single global threshold, mandatory. Deliver Monitoring dashboard with per-site alert bands.	Local versus global anomaly scoring; per-site baseline modelling	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	OpenAQ multi-station and CPCB network data (public) (Find and cite suitable public data yourselves.)	Report which sites benefit most and why.	SMART INDIA HACKATHON 2026 PS ID: SIH26178 A resilient, AI-powered environmental monitoring network that provides early detection, localized intelligence, and actionable alerts for floods, forest fires, pollution events, and other environmental hazards common in India, enabling authorities and communities to shift from reactive disaster response to proactive risk prevention. Organisation: Qualcomm Inc https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G9	Sunaina (12401240) Arun (12417682)	**Seafloor Debris and Anomaly Discovery in Side-Scan Sonar**	Side-scan sonar surveys produce far more seabed imagery than analysts can inspect, and debris is found only where someone happens to look.	Marine survey teams, port and coastal authorities	1. Build a detector that scores sonar tiles by how unlike ordinary seabed they are, ranking them for analyst review. 2. Produce a tile anomaly ranking with precision at top-K against annotated debris. 3. Report precision at top-K, debris recall in the reviewed fraction, review effort saved. Baseline: intensity thresholding, mandatory. Deliver Survey review tool with ranked sonar tiles.	Unsupervised anomaly scoring on sonar image tiles; representation learning	Unit V anomaly detection; Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering	Public side-scan sonar and seabed imagery datasets (Find and cite suitable public data yourselves.)	Cluster the flagged tiles to separate debris types from geological features.	SMART INDIA HACKATHON 2026 PS ID: SIH26057 AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS
G10	Harsh Anand (12400774)	**Unsupervised Condition Monitoring for Unmanned Aerial Propulsion**	Long-endurance missions depend on propulsion reliability, and most airframes have never been run to failure so labelled degradation data does not exist.	UAV operators, maintenance and mission planning teams	1. Build a monitor that learns healthy operation and reports drift away from it, with warning time before recorded failure. 2. Produce a drift detector reporting warning lead time on recorded failures. 3. Report warning lead time, detection rate, false alarms per flight hour. Baseline: fixed parameter limits, mandatory. Deliver Maintenance dashboard with health trend per airframe.	Unsupervised drift detection on telemetry; healthy-baseline modelling	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit III clustering	NASA C-MAPSS turbofan degradation data (public) (Find and cite suitable public data yourselves.)	Discover operating modes first so drift is judged within the correct mode.	SMART INDIA HACKATHON 2026 PS ID: SIH26054 AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs. Organisation: DRDO https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

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G11	Khushbu (12411716)	**Degradation Discovery in Electrical and Electronic Equipment**	Equipment reliability programmes replace parts on a calendar because nobody can see which units are actually degrading.	Utilities, industrial maintenance teams	1. Build a detector that scores each unit by departure from its own healthy operating signature. 2. Produce a per-unit degradation score with lead time against recorded failures. 3. Report lead time, detection rate, false alarms per unit-month. Baseline: calendar replacement schedule, mandatory. Deliver Asset health dashboard with ranked units.	Per-unit anomaly baselines; degradation trend detection	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	SECOM process data and NASA bearing datasets (public) (Find and cite suitable public data yourselves.)	Separate genuine degradation from a change in duty cycle.	SMART INDIA HACKATHON 2026 PS ID: SIH26049 Modifications to improve the reliability, efficiency,and lifespan of electrical and electronic equipment and systems in the ambient condition of subzero temperature and low pressure of High Altitude Areas(HAA) and Super High Altitude Areas (SHAA) of Ladakh region. Organisation: DRDO https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G12	Sanskriti Sharma (12405011)	**Persistent Thermal Source Discovery from Satellite Fire Data**	Thermal satellite feeds flag far more hotspots than can be checked, and a persistent industrial source looks the same as a new wildfire.	Disaster management agencies, pollution control boards	1. Build a system that learns which thermal sources are persistent and normal, so genuinely new events stand out. 2. Produce a novelty detector separating new fire events from persistent industrial sources. 3. Report new-event precision, persistent-source suppression rate, false alarms per day. Baseline: raw hotspot count, mandatory. Deliver Fire monitoring map with new-event highlighting.	Spatio-temporal anomaly detection; persistence modelling	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit III clustering	NASA FIRMS active fire archive (public) (Find and cite suitable public data yourselves.)	Cluster persistent sources to build a reusable industrial-source register.	SMART INDIA HACKATHON 2026 PS ID: SIH26162 AI-Based Detection and Classification of Industrial Fires and Persistent Thermal Sources Using NASA FIRMS, OSM & Satellite Data Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G13	Ayan Kar (12409152)	**Oil Spill Discovery from Satellite Imagery and Vessel Tracks**	Spills at sea are found late and attribution is guesswork, because imagery and vessel movement are never analysed together.	Coast guard, marine pollution authorities	1. Build a detector that flags anomalous dark patches on the sea surface and correlates them with vessel tracks passing through. 2. Produce a spill candidate ranking with vessel correlation, validated against recorded incidents. 3. Report precision at top-K, incident recall, correlation hit rate. Baseline: backscatter thresholding alone, mandatory. Deliver Maritime monitoring map with candidates and tracks.	Unsupervised anomaly detection on imagery; trajectory correlation	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit VI association and sequence mining	Copernicus Sentinel-1 SAR and public AIS vessel tracks (Find and cite suitable public data yourselves.)	Report how many candidate spills have a plausible vessel association.	SMART INDIA HACKATHON 2026 PS ID: SIH26143 Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill. Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G14	Ande Naga Subramanyam (12410806)	**Personal Baseline Stress Monitoring for Uniformed Personnel**	Stress in uniformed services is identified by manual observation and self-report, so deterioration is noticed only once it is severe.	Personnel welfare units, occupational health teams	1. Build a monitor that learns each individual normal physiological and activity rhythm and flags sustained departures. 2. Produce a per-person departure detector with false alarms per week and detection on labelled stress episodes. 3. Report detection rate on labelled episodes, false alarms per person-week, precision at top-K. Baseline: population-level fixed thresholds, mandatory. Deliver Welfare dashboard with per-person trend and flags.	Per-person anomaly baselines on wearable signals; sustained-departure logic	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit III clustering	WESAD and PhysioNet wearable stress datasets (public) (Find and cite suitable public data yourselves.)	Distinguish a single hard day from genuine sustained deterioration.	SMART INDIA HACKATHON 2026 PS ID: SIH26186 AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G15	Anshul (12410144) Vivaan Kumar (12415476)	**Herd-Level Mastitis Signal Discovery in Dairy Records**	Bovine mastitis is confirmed by laboratory test long after milk yield and behaviour have already changed.	Dairy cooperatives, veterinary services, small dairy farmers	1. Build a detector that learns each animal normal production pattern and flags departures before clinical signs appear. 2. Produce a per-animal detector reporting days of warning before recorded diagnosis. 3. Report warning lead time, detection rate, false alarms per herd-month. Baseline: fixed yield-drop threshold, mandatory. Deliver Dairy dashboard with animal and herd alerts.	Per-animal anomaly baselines on production series; early-departure detection	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public dairy herd production and mastitis datasets (Find and cite suitable public data yourselves.)	Aggregate to herd level so a spreading problem is visible.	SMART INDIA HACKATHON 2026 PS ID: SIH26109 AI-Based Predictive Modelling for Early Forecasting of Bovine Mastitis in Indian Dairy Farms Organisation: Ministry of Fisheries, Animal Husbandry & Dairying https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G16	Athrav Sharma (12407534) Abhishek Mor (12401172)	**Tampered Document Discovery Without Labelled Forgeries**	Border and verification desks meet forgery techniques that were never in any training set, so a labelled classifier is always one step behind.	Border control, verification desks, issuing authorities	1. Build a screening system that models genuine documents and scores submissions by how far they depart. 2. Produce a novelty detector evaluated on forgery types held out entirely from training. 3. Report detection on unseen forgery types, false-refusal rate, precision at top-K. Baseline: supervised classifier on known forgeries, mandatory. Deliver Verification desk tool with flagged submissions.	Unsupervised novelty detection on document images; reconstruction-based scoring	Unit V anomaly detection; Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering	Public document forgery datasets (MIDV, DocTamper) (Find and cite suitable public data yourselves.)	Report which tampering types remain invisible to the method.	SMART INDIA HACKATHON 2026 PS ID: SIH26188 AI-Based Fake Identity & Document Screening System Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G17	Arashdip Singh (12408002) Mohd Ismaeel Zaidi (12407497)	**Bottleneck Discovery in Industrial Approval Workflows**	Industrial approvals stall somewhere between departments and nobody can say where, because the process is measured only end to end.	Industry departments, applicant firms, single-window agencies	1. Build a system that scores individual applications by how far their progress departs from the normal path. 2. Produce an anomaly ranking of stalled applications with the responsible stage named. 3. Report precision at top-K stalled cases, stage attribution accuracy, delay explained. Baseline: total-elapsed-time ranking, mandatory. Deliver Approvals dashboard with stalled-case queue.	Process anomaly detection on timestamped workflow events; stage-level baselines	Unit V anomaly detection; Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering	BPI Challenge administrative process event logs (public) (Find and cite suitable public data yourselves.)	Report which stage contributes most total delay across the year.	SMART INDIA HACKATHON 2026 PS ID: SIH26130 Efficiency in streamlining industrial approvals,compliance processes,and access to government support services Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G18	Udit Dewangan (12409130)	**Irregularity Discovery in Public Procurement Participation**	Public procurement is meant to be open to new suppliers, and patterns that quietly exclude them surface only in retrospective audit.	Procurement oversight bodies, startups and MSME bidders	1. Build a detector that scores tenders by how unusual their participation and award pattern is. 2. Produce a tender anomaly ranking with precision on a manually reviewed sample. 3. Report precision at top-K on manual review, flag rate, coverage. Baseline: single-bid-tender rule, mandatory. Deliver Oversight dashboard with flagged tenders.	Unsupervised anomaly detection on tender features; participation pattern modelling	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Open Government Data India procurement and tender datasets (public) (Find and cite suitable public data yourselves.)	Add supplier-network features linking repeat participants.	SMART INDIA HACKATHON 2026 PS ID: SIH26136 Startup friendly public procurement mechanism that enables government departments to identify,pilot, procure and scale innovative solutions from eligible startups Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G19	Sherin James (12410534) Helan Roys (12409127)	**Exposure Anomaly Detection from Colorimetric Dosimeter Readings**	Gas exposure wristbands are read by eye at shift end, so a dangerous spike during the shift leaves no trace.	Industrial safety officers, field workers	1. Build a detector that learns normal exposure profiles per role and flags shifts that depart from them. 2. Produce a shift-level anomaly detector with precision against recorded incident reports. 3. Report precision at top-K shifts, detection on recorded incidents, false alarms per week. Baseline: single fixed exposure limit, mandatory. Deliver Safety dashboard with shift exposure flags.	Per-role anomaly baselines on exposure series; drift detection	Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering; Unit III clustering	Public occupational exposure and air quality sensor datasets (Find and cite suitable public data yourselves.)	Discover role profiles first so exposure is judged in context.	SMART INDIA HACKATHON 2026 PS ID: SIH26118 Passive Colorimetric H2S Exposure-Dosimeter Wristband with AI-Based Quantitative Reading Organisation: Mangalore Refinery and Petrochemicals Limited (MRPL) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G20	Bella Hruday (12407938)	**Cybercrime Complaint Pattern and Outlier Discovery**	Cybercrime complaints are filed and investigated individually, so coordinated campaigns look like unrelated one-off cases.	Cybercrime units, complaint helpdesks	1. Build a system that scores complaints against learned normal patterns and surfaces those indicating a coordinated campaign. 2. Produce an anomaly ranking with burst detection validated against recorded campaigns. 3. Report precision at top-K, campaign detection lead time, false alarms per week. Baseline: raw daily complaint counts, mandatory. Deliver Investigation dashboard with campaign alerts.	Unsupervised anomaly detection on complaint records; temporal burst detection	Unit V anomaly detection; Unit VI association and sequence mining; Unit II preprocessing, scaling and feature engineering	Public cybercrime and consumer complaint datasets (CFPB) (Find and cite suitable public data yourselves.)	Report how early a coordinated campaign becomes visible.	SMART INDIA HACKATHON 2026 PS ID: SIH26184 Development of a Predictive Analytics Framework for Cybercrime Complaints to Forecast Likely Cash Withdrawal Locations in Advance, Enabling Generation of Actionable Intelligence for Timely and Proactive Cybercrime Intervention. Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G21	Dharaneesh Nachimuthu (12405172)	**Audience Structure Discovery in Social Media Communities**	Understanding an online community requires knowing who its members are and what interests them, which a single sentiment score never reveals.	Government communication units, brand analytics teams	1. Build a system that groups community members by behaviour and content into interpretable audience segments. 2. Produce an audience segmentation with stability across resamples and a profile per segment. 3. Report silhouette, cluster stability across resamples, segment separation. Baseline: follower-count quantile grouping, mandatory. Deliver Audience explorer dashboard.	Text and behaviour embedding; audience clustering; segment profiling	Unit III clustering; Unit IV dimensionality reduction; Unit V cluster validity and evaluation	Public Twitter/X and Reddit community datasets (Find and cite suitable public data yourselves.)	Track how segment prevalence shifts over time.	SMART INDIA HACKATHON 2026 PS ID: SIH26152 Social Media Analytics Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

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G22	Raskas Vinay Kumar (12413709)	**Threat Actor Grouping from Dark Web Activity Traces**	Investigators face thousands of pseudonymous handles with no way to tell which belong to the same actor.	Cybercrime investigation units, threat intelligence teams	1. Build a system that groups handles by writing style and behavioural signature without identity labels. 2. Produce a handle clustering with pairwise precision and recall against known linked accounts. 3. Report pairwise F1, cluster purity, linked-account recall. Baseline: exact username matching, mandatory. Deliver Investigation tool with proposed actor groups.	Stylometric feature extraction; identity clustering; similarity thresholding	Unit III clustering; Unit IV dimensionality reduction; Unit V cluster validity and evaluation	PAN author identification corpora (public) (Find and cite suitable public data yourselves.)	Handle actors who deliberately vary their style.	SMART INDIA HACKATHON 2026 PS ID: SIH26151 Dark web threat actor de-anonymization Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G23	Geethu Rajesh (12412926)	**Criminal Network Community Discovery from Case Records**	Investigators see individual links between people but not the groups those links form.	Police investigation units, crime analysts	1. Build a system that builds a network from case associations and discovers its communities without labels. 2. Produce a community structure with modularity and stability, plus profiles of the largest groups. 3. Report modularity, community stability across resamples, purity against known groups. Baseline: connected components, mandatory. Deliver Network explorer with community overlay.	Graph construction; community detection; cluster validity on graphs	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public criminal network and co-offending datasets (Find and cite suitable public data yourselves.)	Detect individuals bridging two communities.	SMART INDIA HACKATHON 2026 PS ID: SIH26189 AI-Powered Criminal Network Analysis System Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G24	Bheemanapalli Vijaya Shanmukha Sai (12409607) Mandli Jagannmohan Reddy (12402589)	**Cryptographic Asset Inventory Grouping Across an Enterprise**	Enterprises cannot answer what cryptography they are running, because certificates and keys accumulate with no inventory.	Enterprise security architects, compliance teams	1. Build a system that groups discovered cryptographic assets by configuration profile to expose weak and outdated families. 2. Produce an asset clustering with profiles per group and the weak families identified. 3. Report silhouette, cluster purity against algorithm families, coverage. Baseline: algorithm-name grouping, mandatory. Deliver Inventory dashboard with asset group profiles.	Configuration feature engineering; clustering of asset profiles	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Censys and Rapid7 open TLS certificate scans (public) (Find and cite suitable public data yourselves.)	Flag assets matching no established group as candidate shadow deployments.	SMART INDIA HACKATHON 2026 PS ID: SIH26164 Enterprise Cryptographic Discovery & Analysis Tool (ECDAT) Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G25	Peddireddy Gari Pramod kumar (12402043)	**Device Configuration Grouping Across a Multi-Vendor Network**	Networks run hardware from many vendors whose configurations are compared against standards one device at a time.	Network security teams, compliance auditors	1. Build a system that groups device configurations by behaviour to reveal how many genuinely distinct postures exist. 2. Produce a configuration clustering with the distinct postures named and outliers flagged. 3. Report silhouette, cluster purity against vendor, outlier precision. Baseline: vendor-based grouping, mandatory. Deliver Compliance dashboard with posture groups.	Configuration feature extraction; clustering; outlier configuration detection	Unit III clustering; Unit V anomaly detection; Unit II preprocessing, scaling and feature engineering	Public network device configuration corpora (Find and cite suitable public data yourselves.)	Report which cluster is furthest from the applicable benchmark.	SMART INDIA HACKATHON 2026 PS ID: SIH26155 AI-Driven Multi-Vendor Network Security Compliance Auditor Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G26	Aryan (12408182) Tanvir Kaur Randhawa (12408255)	**VPN Session Behaviour Grouping for Security Assessment**	Tunnel configurations are reviewed individually, so systemic weak patterns across the estate are never seen.	Network security teams, infrastructure operators	1. Build a system that groups VPN sessions by negotiated parameters and behaviour to expose recurring weak patterns. 2. Produce a session clustering with a weakness profile per group. 3. Report silhouette, purity against known configuration classes, coverage. Baseline: cipher-suite string grouping, mandatory. Deliver Assessment dashboard with session groups.	Session feature engineering; clustering; weak-pattern profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public network traffic captures containing IPsec sessions (Find and cite suitable public data yourselves.)	Track whether weak groups shrink after remediation.	SMART INDIA HACKATHON 2026 PS ID: SIH26160 AI-Powered IPsec VPN Protocol Analyzer and Security Assessment Framework Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G27	Moumita Maji (12403919)	**Email Security Posture Grouping Across Domains**	Every organisation configures mail security differently, and reviewing domains one at a time hides the common failure patterns.	Email administrators, sector security bodies	1. Build a system that groups domains by security posture to reveal the recurring configurations that leave them exposed. 2. Produce a domain posture clustering with the exposure profile of each group. 3. Report silhouette, cluster stability, exposure separation between groups. Baseline: single compliance-score quantiles, mandatory. Deliver Sector dashboard with posture groups.	Posture feature extraction; clustering; profile interpretation	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public DNS, SPF and DMARC scan datasets (Find and cite suitable public data yourselves.)	Report how posture groups differ by organisation type.	SMART INDIA HACKATHON 2026 PS ID: SIH26159 SecureMailScope: AI-Assisted Cryptographic Security Posture Assessment for Secure Email Communications Organisation: National Technical Research Organisation (NTRO) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G28	Bommanaboyina Rudra Veera Narayana (12400377)	**Wallet Clustering for Virtual Asset Service Attribution**	Unknown wallets must be attributed to the service controlling them, and no labelled mapping exists for most of them.	Financial intelligence units, exchange compliance teams	1. Build a system that groups wallets by transactional behaviour so unknown wallets inherit the profile of their group. 2. Produce a wallet clustering with attribution accuracy against known service labels held out. 3. Report cluster purity, attribution accuracy on held-out labels, adjusted Rand index. Baseline: transaction-volume quantile grouping, mandatory. Deliver Attribution tool with wallet group profiles.	Behavioural feature engineering; wallet clustering; nearest-group attribution	Unit III clustering; Unit IV dimensionality reduction; Unit V cluster validity and evaluation	Elliptic Bitcoin dataset and public exchange-labelled address sets (Find and cite suitable public data yourselves.)	Report confidence so weak attributions are not acted on.	SMART INDIA HACKATHON 2026 PS ID: SIH26182 Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G29	Devesh Prasad Tiwari (12409087)	**Scene Structure Discovery in Border Surveillance Footage**	Existing CCTV estates record continuously and are reviewed only after an incident, so ordinary versus unusual activity is never characterised.	Border security forces, surveillance control rooms	1. Build a system that groups recorded scenes by activity signature so reviewers see one representative per group. 2. Produce a scene clustering with purity against annotated activity and the review effort saved. 3. Report cluster purity, adjusted Rand index, review reduction. Baseline: motion-energy thresholding, mandatory. Deliver Review interface with one clip per scene group.	Video feature extraction; scene clustering; representative sampling	Unit III clustering; Unit IV dimensionality reduction; Unit II preprocessing, scaling and feature engineering	UCF-Crime and VIRAT surveillance datasets (public) (Find and cite suitable public data yourselves.)	Flag scenes matching no established group for priority review.	SMART INDIA HACKATHON 2026 PS ID: SIH26187 AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure. Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G30	Ediga Rushyanth (12410406)	**Weather Regime Discovery in a National Big Data Archive**	National weather archives hold decades of observations queried by date rather than by the kind of weather they contain.	Meteorological departments, climate researchers	1. Build a system that discovers recurring weather regimes from the archive so days can be retrieved by behaviour. 2. Produce a regime clustering with stability across years and a named profile per regime. 3. Report silhouette, regime stability across held-out years, separation. Baseline: calendar-season grouping, mandatory. Deliver Archive explorer searchable by regime.	Multivariate time-series clustering; regime discovery; validity testing	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	IMD gridded observations and ERA5 reanalysis (public) (Find and cite suitable public data yourselves.)	Test whether regime frequency has shifted across decades.	SMART INDIA HACKATHON 2026 PS ID: SIH26069 National Weather Big Data Analytics Platform Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G31	Syed Afzal Syed Afzan Farsi (12404167)	**Forecast Behaviour Grouping for Multi-Model Blending**	Forecast models are blended with fixed weights although each performs differently depending on the situation.	Meteorological forecasting centres	1. Build a system that discovers the situation types in which each model behaves differently, so blending can be conditioned on them. 2. Produce a situation clustering showing model skill differs significantly between clusters. 3. Report silhouette, between-cluster skill separation, stability. Baseline: single global blend weighting, mandatory. Deliver Forecaster dashboard with situation groups.	Clustering of forecast-situation features; per-cluster model behaviour profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public NWP ensemble archives and IMD observations (Find and cite suitable public data yourselves.)	Derive per-cluster blend weights and evaluate them.	SMART INDIA HACKATHON 2026 PS ID: SIH26081 Hybrid AI–NWP Multi-Model Forecast Blending System Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G32	Rajana Praveen (12406920)	**Heat Stress Regime Discovery for Early Warning**	Heat warnings use a single temperature threshold although the combinations of heat, humidity and wind that harm people differ by place.	Public health departments, urban local bodies	1. Build a system that discovers distinct heat stress regimes and characterises which population each endangers. 2. Produce a heat regime clustering with health-outcome association per regime. 3. Report silhouette, outcome separation between regimes, stability. Baseline: fixed temperature threshold, mandatory. Deliver Heat advisory dashboard with regime map.	Multivariate clustering on meteorological and stress indices; regime profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	IMD temperature and humidity series with public heat mortality data (Find and cite suitable public data yourselves.)	Report regimes a single temperature threshold entirely misses.	SMART INDIA HACKATHON 2026 PS ID: SIH26083 Extreme Heatwave Early Warning and Human Thermal Stress Index Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G33	Devulapally Abhiram (12405768)	**Rainfall and Drainage Regime Discovery for Urban Flood Nowcasting**	Urban flooding depends on how rainfall meets drainage capacity, and the recurring combinations causing it are never enumerated.	Municipal drainage departments, city disaster cells	1. Build a system that discovers the recurring rainfall-drainage regimes under which particular wards flood. 2. Produce a regime clustering with per-ward flooding association. 3. Report silhouette, flood-rate separation between regimes, stability. Baseline: rainfall-total quantile grouping, mandatory. Deliver City dashboard with ward risk by regime.	Coupled multivariate clustering; regime profiling by ward	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	IMD rainfall data with public urban flood incident reports (Find and cite suitable public data yourselves.)	Identify wards that flood under regimes considered mild elsewhere.	SMART INDIA HACKATHON 2026 PS ID: SIH26085 Urban Flood Nowcasting System (Drainage and Rainfall Coupling) Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

Group	Students	Project Title	Problem to Solve	Who Benefits	Objectives (what you will be assessed on)	Suggested Techniques	Syllabus Topics Used	Data to Use	Optional Advanced Extension	Inspired By (official source)
G34	Nakkanaboina Surendra (12405682)	**Catchment Behaviour Grouping for Flash Flood Risk in Hills**	Flash flood warnings treat hill catchments alike although their response to the same rainfall differs enormously.	State disaster authorities, hill district administrations	1. Build a system that groups catchments by hydrological response so warnings are issued per group rather than uniformly. 2. Produce a catchment clustering with response profiles validated against recorded flash floods. 3. Report silhouette, response separation, event association. Baseline: elevation-band grouping, mandatory. Deliver Regional dashboard with catchment groups.	Catchment feature engineering; response-curve clustering	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public catchment, DEM and rainfall-runoff datasets (Find and cite suitable public data yourselves.)	Report which catchments respond fastest and need the earliest warning.	SMART INDIA HACKATHON 2026 PS ID: SIH26192 Flash Flood Prediction System for Hilly Regions using Multi-Source Data Theme Organisation: Ministry of Home Affairs https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G35	Umyal Dixit (12417102)	**Slope Condition Grouping for Landslide Risk Monitoring**	Landslide monitoring in the North East is reactive, and vulnerable slopes are identified only after failure.	State disaster authorities, road and highway agencies	1. Build a system that groups slope segments by terrain, rainfall and soil signature so risk can be assigned by group. 2. Produce a slope clustering with historical landslide rate reported per group. 3. Report silhouette, landslide-rate separation, stability. Baseline: slope-angle threshold grouping, mandatory. Deliver Risk map with slope group overlay.	Geospatial feature engineering; slope-segment clustering; profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Bhuvan terrain layers, public landslide inventories, IMD rainfall (Find and cite suitable public data yourselves.)	Flag segments sitting far from any established group.	SMART INDIA HACKATHON 2026 PS ID: SIH26001 AI-Based early warning and landslide Risk Monitoring System in NER Organisation: Ministry of Development of North Eastern Region (MDoNER) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G36	Padala Sai Srikanth (12405956)	**Spatial Micro-Climate Grouping from Block to Panchayat Scale**	Weather advisories are issued at block level while conditions vary sharply between panchayats within a block.	Agro-advisory services, panchayat administrations, farmers	1. Build a system that groups panchayats by observed micro-climate behaviour so advisories are issued per group. 2. Produce a panchayat clustering with within-block variation quantified. 3. Report silhouette, within-block variance explained, stability. Baseline: block-level uniform grouping, mandatory. Deliver Advisory map with micro-climate groups.	Spatial clustering on multivariate climate series; regime profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	IMD gridded data and public agro-meteorological station series (Find and cite suitable public data yourselves.)	Report blocks where a single advisory is clearly inadequate.	SMART INDIA HACKATHON 2026 PS ID: SIH26074 Downscaling of weather forecast from Block level to Panchayat level: Inferring high-resolution plots/ data/ information from low-resolution plot /data /information /variables for agro-meteorological advisory services. Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G37	Ashwell Cherian Giji (12408107)	**Monsoon Onset Pattern Discovery at Village Scale**	Sowing decisions depend on monsoon onset, which is declared regionally although it varies village to village.	Farmers, agricultural extension services	1. Build a system that discovers recurring local onset patterns so sowing advice reflects the pattern a village actually experiences. 2. Produce an onset pattern clustering with year-to-year stability per location. 3. Report silhouette, pattern stability across years, separation. Baseline: regional declared onset date, mandatory. Deliver Sowing advisory map by onset pattern.	Time-series shape clustering on rainfall onset curves	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	IMD gridded rainfall and public district rainfall series (Find and cite suitable public data yourselves.)	Report locations whose onset pattern has shifted over the record.	SMART INDIA HACKATHON 2026 PS ID: SIH26086 Hyperlocal Monsoon Onset & Break Prediction System (Block/Village Scale) Organisation: Ministry of Earth Sciences (MoES) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G38	Rithika Kotra (12407476)	**Produce Storage Demand Grouping for Rural Cold Chain Siting**	Cold storage is sited where land is available rather than where produce needs it, so capacity sits idle while crops spoil elsewhere.	Cold chain operators, farmer producer organisations	1. Build a system that groups localities by produce mix and spoilage-risk profile to guide where storage belongs. 2. Produce a locality clustering with a storage-need profile per group. 3. Report silhouette, need separation between groups, stability. Baseline: total-production ranking, mandatory. Deliver Siting dashboard with need map.	Multivariate clustering on production and seasonality features; profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Agricultural Census, Horticulture Statistics and APMC arrivals (public) (Find and cite suitable public data yourselves.)	Compare the discovered need map against existing cold storage locations.	SMART INDIA HACKATHON 2026 PS ID: SIH26005 Solar-Powered Smart Mini Cold Storage System for Fresh Vegetables in North Eastern Region (NER) Organisation: Ministry of Development of North Eastern Region (MDoNER) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G39	Gulluru Gowtham Reddy (12403114)	**Agricultural Parcel Grouping for Survey Prioritisation**	Rural land resurvey is a vast undertaking and parcels are visited in administrative order rather than by need.	Revenue departments, survey teams, landowners	1. Build a system that groups parcels by land-use and record-consistency signature so survey effort targets the groups that need it. 2. Produce a parcel clustering with a record-discrepancy profile per group. 3. Report silhouette, discrepancy separation, stability. Baseline: village-order survey sequence, mandatory. Deliver Survey planning map with parcel groups.	Geospatial and record feature engineering; parcel clustering	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Bhuvan land use layers and public cadastral datasets (Find and cite suitable public data yourselves.)	Rank groups by expected survey benefit.	SMART INDIA HACKATHON 2026 PS ID: SIH26010 Survey/Resurvey of Rural Agricultural Land in India Organisation: Ministry of Rural Development https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G40	Rahul Rathore (12407470)	**Market Intermediation Structure Discovery in Agricultural Trade**	Multiple intermediaries cut farmer earnings and raise consumer prices, but the actual chain structure is never mapped.	Farmer producer organisations, agricultural marketing boards	1. Build a system that groups markets by intermediation and price-spread signature to expose where the chain is longest. 2. Produce a market clustering with the farmer-to-consumer spread reported per group. 3. Report silhouette, spread separation between groups, stability. Baseline: state-wise grouping, mandatory. Deliver Market structure dashboard.	Clustering on price-spread and flow features; market profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Agmarknet mandi price series and public APMC data (Find and cite suitable public data yourselves.)	Relate the discovered groups to distance and infrastructure.	SMART INDIA HACKATHON 2026 PS ID: SIH26033 Multiple intermediaries reduce farmers earnings and increase consumer prices. Organisation: Ministry of Consumer Affairs, Food & Public Distribution https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G41	Saket Kumar Jha (12411317) Aman Choudhary (12410322)	**Price Behaviour Regime Discovery for Farmer Market Decisions**	Farmers sell immediately after harvest because they cannot tell whether the current price is normal for this market at this time.	Smallholder farmers, producer groups, market boards	1. Build a system that discovers price-behaviour regimes per commodity so a farmer sees which regime the market is currently in. 2. Produce a price regime clustering with the historical outcome of selling in each regime. 3. Report silhouette, regime separation in realised price, stability across years. Baseline: monthly average price, mandatory. Deliver Farmer-facing dashboard showing current regime.	Time-series regime clustering on price series; regime characterisation	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Agmarknet daily mandi price series (public) (Find and cite suitable public data yourselves.)	Detect regime transitions as they begin rather than after the fact.	SMART INDIA HACKATHON 2026 PS ID: SIH26132 Strengthening market linkages and price discovery for farmers Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G42	Vipin Soni (12407728)	**Feed and Silage Quality Grouping for Dairy Advisory**	Feed quality is assessed by laboratory tests most small dairy farmers never commission, so poor feed is diagnosed through lost yield.	Dairy cooperatives, small dairy farmers, veterinary advisors	1. Build a system that groups feed samples by measured signature so a farmer is told which quality group their feed belongs to. 2. Produce a feed quality clustering with purity against laboratory grades held out. 3. Report cluster purity, adjusted Rand index against grades, silhouette. Baseline: single moisture-content threshold, mandatory. Deliver Advisory tool returning feed quality group.	Spectral and compositional feature engineering; sample clustering	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public feed composition and NIR spectral datasets (Find and cite suitable public data yourselves.)	Flag samples unlike any established group for laboratory referral.	SMART INDIA HACKATHON 2026 PS ID: SIH26111 Smart AI-Enabled Rapid Feed and Silage Quality Testing System for Dairy Farmers Organisation: Ministry of Fisheries, Animal Husbandry & Dairying https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G43	Nishant Prajapati (12417878)	**Micro-Enterprise Profile Discovery for Rural Business Advisory**	Rural micro-entrepreneurs receive identical generic advice because nobody has characterised the distinct kinds of business that exist.	Rural livelihood missions, micro-entrepreneurs, lending institutions	1. Build a system that groups micro-enterprises by operating and financial signature so advisory is targeted per group. 2. Produce an enterprise clustering with a needs profile and prevalence per group. 3. Report silhouette, outcome separation, stability under imputation choices. Baseline: turnover quantile grouping, mandatory. Deliver Advisory dashboard with enterprise groups.	Mixed-type clustering; missing-data handling; group profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	MSME Census and public rural enterprise survey data (Find and cite suitable public data yourselves.)	Test whether the groups differ in credit outcomes.	SMART INDIA HACKATHON 2026 PS ID: SIH26091 AI-Driven Hyper-Local Business Advisory and Financial Structuring Assistant for Rural Micro-Entrepreneurs Organisation: Ministry of Social Justice and Empowerment (MoSJE) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G44	Akhil Lankalapalli (12406138) Esuru Narendranath (12413472)	**Livelihood Pattern Discovery for Skilling Recommendations**	Skilling programmes are offered by qualification category while rural livelihoods combine several activities no category describes.	Skill development missions, rural households	1. Build a system that groups households by livelihood activity mix so skilling matches how people actually earn. 2. Produce a livelihood clustering with an activity profile and skilling implication per group. 3. Report silhouette, activity separation, stability. Baseline: primary-occupation grouping, mandatory. Deliver Skilling planning dashboard.	Mixed-type clustering on activity profiles; group profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Periodic Labour Force Survey and NSSO employment data (public) (Find and cite suitable public data yourselves.)	Report which groups the existing qualification categories merge wrongly.	SMART INDIA HACKATHON 2026 PS ID: SIH26097 AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY Organisation: Ministry of Social Justice and Empowerment (MoSJE) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data

Group	Students	Project Title	Problem to Solve	Who Benefits	Objectives (what you will be assessed on)	Suggested Techniques	Syllabus Topics Used	Data to Use	Optional Advanced Extension	Inspired By (official source)
G45	Saranjeet Singh (12408620) Vemulapalli Nithin Ramachandra (12406760)	**Healthcare Access Barrier Grouping in Underserved Districts**	Rural health programmes are designed for an average patient who does not exist, because access barriers cluster into distinct patterns.	District health administrations, primary health centres	1. Build a system that groups households by access barrier pattern so each group receives the intervention it needs. 2. Produce a barrier pattern clustering with prevalence and a needs profile per group. 3. Report silhouette, stability across imputation choices, profile separation. Baseline: distance-to-facility stratification, mandatory. Deliver Programme planning dashboard.	Mixed-type clustering; missing-data handling; barrier profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	NFHS and District Level Household Survey data (public) (Find and cite suitable public data yourselves.)	Test stability under different missing-data strategies.	SMART INDIA HACKATHON 2026 PS ID: SIH26133 Accessibility and quality of public healthcare services,particularly in rural and underserved areas Organisation: Government Of Maharashtra https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G46	Sachin Kumar (12400346) Jitendra Choudhary (12404996)	**Presentation Pattern Discovery in Structured Patient Histories**	History taking is the most important diagnostic step, and the recurring presentation patterns within it are never characterised across patients.	Hospital clinicians, medical educators	1. Build a system that groups patient histories into recurring presentation patterns to support structured case taking. 2. Produce a presentation clustering with purity against recorded diagnosis groups held out. 3. Report cluster purity, adjusted Rand index, silhouette. Baseline: chief-complaint grouping, mandatory. Deliver Case-taking tool showing the matched pattern.	Mixed-type clustering on structured history fields; pattern profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	MIMIC-IV structured clinical data (public, credentialed) (Find and cite suitable public data yourselves.)	Report patients whose presentation fits no established pattern.	SMART INDIA HACKATHON 2026 PS ID: SIH26047 Patient Case-Taking Software Organisation: Ministry of Ayush https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G47	Priyam Kumari (12407782) Aradala Pawan Kalyan (12402723)	**Osteoarthritis Risk Marker Grouping for Early Screening**	Osteoarthritis is diagnosed once symptomatic, although measurable risk markers combine into patterns long before that.	Orthopaedic services, community screening programmes	1. Build a system that groups individuals by risk marker signature so screening targets the highest-risk groups. 2. Produce a risk marker clustering with outcome rates reported per group. 3. Report silhouette, outcome separation, stability. Baseline: age-band stratification, mandatory. Deliver Screening dashboard with risk groups.	Multivariate clustering on risk markers; group profiling and validation	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Osteoarthritis Initiative (OAI) public cohort data (Find and cite suitable public data yourselves.)	Test whether the groups reproduce age alone or add genuine information.	SMART INDIA HACKATHON 2026 PS ID: SIH26004 AI-Assisted Early Detection System for Osteoarthritis (OA) Risk Markers in North Eastern Region (NER) Organisation: Ministry of Development of North Eastern Region (MDoNER) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G48	Nishar (12402674) Aakriti Sharma (12407752)	**Distress Trajectory Discovery in Mental Health Monitoring**	Mental health support treats distress as a single score although people follow distinct trajectories calling for different responses.	Mental health services, victim support programmes	1. Build a system that groups individuals by distress trajectory shape so support matches the trajectory rather than the current level. 2. Produce a trajectory clustering with outcome differences reported per group. 3. Report silhouette, outcome separation, earliest separable timepoint. Baseline: current-score quantile grouping, mandatory. Deliver Care dashboard with trajectory groups.	Time-series trajectory clustering; shape feature extraction	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public longitudinal mental health survey datasets (Find and cite suitable public data yourselves.)	Identify the earliest point at which trajectories become distinguishable.	SMART INDIA HACKATHON 2026 PS ID: SIH26094 AI-Powered Dynamic Mental Health Monitoring and Distress Prediction System for Victims of Atrocities Organisation: Ministry of Social Justice and Empowerment (MoSJE) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G49	Mohammed Safran K (12414449)	**Cognitive Engagement Pattern Discovery in Dementia Care Games**	Cognitive games record every interaction and report only a score, discarding the engagement patterns that indicate change.	Geriatric care providers, families, dementia researchers	1. Build a system that groups play sessions by interaction pattern so carers see behavioural change rather than a score. 2. Produce a session clustering with per-person pattern profiles over time. 3. Report silhouette, within-person consistency, shift detection precision. Baseline: total score trend, mandatory. Deliver Carer dashboard with engagement patterns.	Interaction feature extraction; session clustering; per-person profiling	Unit III clustering; Unit II preprocessing, scaling and feature engineering; Unit V cluster validity and evaluation	Public cognitive assessment and interaction log datasets (Find and cite suitable public data yourselves.)	Detect the point at which an individual pattern shifts group.	SMART INDIA HACKATHON 2026 PS ID: SIH26003 AI-Based Cognitive Gaming and Memory Assistance Platform for Elderly Dementia Patients in North Eastern Region (NER) Organisation: Ministry of Development of North Eastern Region (MDoNER) https://sih.gov.in/sih2026PS Dataset access: Student-sourced public data
G50	Saathwik manthri (12321227)									